

When Do Curricula Work?

Motivation. Curriculum learning — presenting training examples in a meaningful order, typically easy-to-hard — has been proposed as a way to improve convergence and generalization. Anti-curriculum learning (hard-to-easy) has also shown benefits in some settings. Despite a rich literature, empirical results conflict: curricula help in some experiments but not others, and no ordered learning method consistently improves across contexts. This paper aims to systematically determine when and why curricula are beneficial.

Implicit Curricula. Before studying explicit orderings, the authors investigate whether neural networks already learn examples in a consistent order under standard i.i.d. training. They define the *learned iteration* of a sample as the epoch at which the model first correctly and permanently classifies it.

Across 142 configurations (architectures $\times$ optimizers) on CIFAR-10, the learned iteration ordering is highly consistent within architecture families. The Spearman correlation between orderings from different CNNs is very high, indicating that a robust, architecture-dependent notion of example difficulty exists. This *implicit curriculum* emerges from the interaction of architecture and optimization, not from explicit data ordering.

Curriculum Framework. The paper formalizes curricula via three components:

- (a) **Scoring function** $s(\mathbf{x}) \in \mathbb{R}$: assigns a difficulty score to each example. Three are studied: loss of a reference model, learned iteration, and c-score (consistency of correct predictions across independent training runs).
- (b) **Pacing function** $g(t)$: determines how many examples are available at training step t , controlling how quickly the full dataset is introduced. Six families are considered: logarithmic, exponential, step, linear, quadratic, and root.
- (c) **Order**: curriculum (easy-to-hard), anti-curriculum (hard-to-easy), or random ordering within the paced subset.

All three scoring functions are highly correlated, so c-score is used throughout.

Main Finding: Curricula Provide Marginal Benefit in Standard Settings. Across a massive sweep of over 25,000 models on CIFAR-10, CIFAR-100, FOOD101, and FOOD101N, the paper finds:

- Under standard training (full budget, clean data), **no ordering — curriculum, anti-curriculum, or random — provides statistically significant improvement** over standard i.i.d. SGD.
- Any marginal gains from pacing functions are explained entirely by the *dynamic dataset size* effect (starting with a smaller dataset and growing it), not by the ordering itself. Random-curriculum performs as well as curriculum.

When Curricula Do Help. Curricula become genuinely beneficial in two specific regimes:

- **Limited training time:** When the compute budget is drastically reduced (e.g., 352 steps instead of 17,600), curriculum ordering provides a clear advantage over anti-curriculum and standard training. At the tightest budgets, even random-curriculum (random order with a pacing function) outperforms the baseline, suggesting the dynamic dataset size itself helps.
- **Noisy labels:** With 20–80% label noise on CIFAR-100, curriculum learning significantly outperforms all other methods. The easy-to-hard ordering allows the model to first learn clean patterns before encountering corrupted examples. The best pacing functions under noise correspond to simply ignoring all noisy data during training.

These findings generalize to larger-scale datasets (FOOD101, FOOD101N).

Why These Regimes. Both limited time and label noise shift the c-score distribution toward harder examples, making the distinction between easy and hard examples more consequential. In standard settings with sufficient compute and clean data, SGD eventually learns all examples regardless of order, so curricula add nothing. When resources are constrained or data is corrupted, prioritizing reliable examples first becomes valuable.

Limitations. The study is limited to image classification with fixed datasets (CIFAR, FOOD101) and does not cover language modeling, where curricula are standard practice in large-scale pretraining (GPT-3, T5). The scoring functions are computed before training begins, so self-paced or online difficulty estimation is not considered. The paper also notes that conclusions about curricula may not transfer to very different task types or training regimes.

Takeaway. Curricula are not a general-purpose improvement to training. Their benefit is concentrated in settings where learning resources are scarce (limited compute or corrupted data), making example prioritization valuable. In standard well-resourced training, the order of examples is largely irrelevant — i.i.d. sampling works just as well.